

Computational Viability Certification under Incomplete Observation: A Continuous-to-Finite Bridge with Exact Certificates, a Solver-Assisted Certified Campaign, and a Finite-Side Reference Library

Amin Abaee

Independent Researcher

ORCID: 0000-0002-0019-1842

amin_abaee@ut.ac.ir

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Abstract

Nonviability certificates are useful computationally only if they are finite, attainable by standard solvers, and independently checkable. This paper develops such a pipeline for linear systems under exogenous partial observation of a hidden mode. An outer moment approximation of all measurable information-adapted controls, combined with an inner adversarial approximation (finitely stored admissible labels and scenarios), yields a finite linear program whose optimum is a certified lower bound on the continuous-time safety value and whose error inflation is a certified upper bound ($\rho_{\mathcal{H},G} \leq J_{\mathcal{I}}(T) \leq \rho_{\mathcal{H},G} + \bar{e} + \delta_{\beta} + Lh_t$, $\bar{e}$ the program’s own verified moment-error bound, $R_U V_U h_u$ in the uniform choice); every dual-feasible solution yields a finite lower-bound certificate covering every measurable policy, and a strictly positive certified margin is a finite obstruction certificate; along every admissible refinement sequence whose mesh and stored-scenario/enclosure widths vanish, nonviability is eventually detected, and whenever a positive certificate exists one can be chosen with at most $r + 1$ safety labels at rank r — the relevant dimension being information–time rank, not instantaneous input dimension, and a matching construction with scalar input produces obstructions whose minimal certificates require arbitrarily many witnesses, saturating the $r + 1$ bound. The certificates account for post-observation recourse: on the three-branch delayed-observation instance every pair of compatible states is viable (explicit pairwise policies are carried exactly) and a common control keeps every branch safe throughout the blind window, yet nonviability is certified, with the exact delay threshold $\tau^* = 7/50$; the certificate decodes into an exact shared-authority delay hierarchy — the triple is obstructed past $\tau = 7/50$, the pairs past ≈ 0.245 and ≈ 0.397 , singletons never — and the printed blind-window controls are exactly the optimal shared-braking programs. The certificate is prescriptive beyond the verdict: the audited $a = 1$ observation-time margin changes at unit rate; post-revelation authority plots are algebraic expressions, not proved optimized delay sensitivities. Belief-cell rows yield conditional one-sided obstructions; an upper sandwich and two-sided margins require separately verified full label and held-policy errors. The finite program is solved by a standard solver (HiGHS) across a five-point mesh refinement; optimality

is certified in exact rational arithmetic by independent primal and dual witnesses, the solver agreeing with the exact values to at most 4.2×10^{-17} , and the refinement obeys the closed law $\rho(h) = 3/50 - Th/4$ (an unrestricted level, nonnegative on this family for $h \leq \frac{1}{5}$). A finite-side reference library, **viacert**, packages the exact finite-horizon recursion and the fibre-partition criteria behind a plug-gable finite-system interface, with twelve audited identities in its self-test; the bridge’s continuous programs live in the campaign scripts, not in the library. All exact verification paths for the reported rational benchmark run in integer and rational arithmetic; solver outputs are numerical confirmations, never proof objects.

Keywords: viability; incomplete observation; obstruction certificates; linear programming; delayed observation; exact arithmetic; certified computation

1. Introduction

Viability theory under full observation has a mature numerical apparatus: kernel approximations, set-valued meshes, and convergence results (Aubin, 1991; Cardaliaguet, Quincampoix, and Saint-Pierre, 1999). Under incomplete observation the sufficiency side is carried by output-feedback conditions and the estimation-tube program (Veliov, 1993), while the necessity side — certificates whose firing *proves* nonviability of an information state (Abaee, 2026, An obstruction calculus) — has lacked a computational bridge. The difficulty is the quantifier order: the adversary chooses the mode, the initial state inside its cell, and a measurable disturbance with knowledge of the announced policy, so naive discretization of controls proves at most the infeasibility of a held-control subclass, not of the policy class itself.

This paper builds the bridge and the tooling around it. The main result (Theorem 4) approximates the admissible controls from outside by their block moments — a necessary relaxation of continuous-time safety that restricts nothing — and the adversary from inside by explicitly stored scenarios, yielding a finite linear program whose optimal value sandwiches the exact minimax safety violation. Dual solutions of the program are finite, policy-independent obstruction certificates; mesh refinement eventually certifies — on a nonviable system sufficiently fine meshes produce a positive certificate, though

positivity is not itself monotone under refinement (Theorem 4(iii)); and their size is governed by the rank of the safety-row matrix in the information–time product, which no bound on the instantaneous input dimension controls (Proposition 5).

The pipeline is demonstrated end to end on a three-branch delayed-observation instance (the running instance of this paper; Section 6): delay threshold, Farkas certificate, and a five-point mesh campaign in which the HiGHS solver (Huangfu and Hall, 2018) agrees with the exact rational optimum at every mesh to floating-point tolerance, with independent primal–dual witnesses closing the duality gap exactly. The finite-side machinery — the exact backward recursion and the fibre-partition criteria — is packaged as the reference library `viacert` with a pluggable finite-system interface and a self-test of twelve audited identities; the library supplies the enumerate-and-check layer of the certification workflow, and it is deliberately not a reimplementaion of the bridge’s continuous programs (Section 11).

The contributions are: the continuous-to-finite sandwich theorem with sound, refinement-complete, sparsity-bounded certificates (Section 5); the recourse-aware obstruction reading and the exact three-branch instance with its certified campaign (Section 6); the information–time rank separation (Section 8); the refinement-erosion identity, the oracle-recourse extension, and the belief-cell certificates for partial and noisy observation (Section 9); the certificate’s prescriptive decode into optimal shared braking authorities, the induced pair-delay hierarchy, and the scoped redesign witness (Section 10); the complexity and residual-bounded verification framework (Section 7); and the reference library (Section 11).

2. Model, information, and quantifiers

Dynamics. Let the hidden mode be $\theta \in \Theta := \{1, \dots, M\}$. Conditional on θ , the continuous state satisfies

$$\dot{x}(t) = A_\theta(t)x(t) + g_\theta(t) + B_\theta(t)u(t) + E_\theta(t)d(t), \quad t \in [0, T],$$

under the standing assumptions: $A_\theta, g_\theta, B_\theta, E_\theta$ bounded and piecewise continuous, with bounded-variation bounds on B_θ, E_θ — the operative assumption below is the induced bound $\text{TV}(k_a) \leq V_U$ on the adjoint kernels, assumed directly in Theorem 4 rather than derived from the primitive coefficients here; the initial set in mode θ a nonempty compact polytope $P_\theta \subset \mathbb{R}^n$; the input and disturbance sets nonempty compact polytopes $U = \{v : Fv \leq f\} \subset \mathbb{R}^m$ and $D_\theta \subset \mathbb{R}^p$; safety the closed polyhedron $\mathcal{V} = \{x : c_i^\top x \leq b_i, i = 1, \dots, s\}$; and the initial information state $B_0 = \bigcup_\theta \{\theta\} \times P_\theta$ initially safe, meaning $P_\theta \subseteq \mathcal{V}$ for every θ (if some P_θ leaves $\mathcal{V}$ at time zero, a time-zero certificate suffices and the analysis is vacuous). The dynamics admit unique Carathéodory solutions for every admissible measurable signal pair.

Information. At known times $0 = \tau_0 < \tau_1 < \dots < \tau_R = T$ the controller receives exogenous information about θ : refining history partitions $\mathcal{P}_0 \preceq \mathcal{P}_1 \preceq \dots \preceq \mathcal{P}_{R-1}$ of Θ — the order is by refinement: for $r' > r$ every cell of $\mathcal{P}_{r'}$ lies inside a cell of $\mathcal{P}_r$ (the last partition is $\mathcal{P}_{R-1}$ because $\tau_R = T$ ends the last interval) — the observation history during $I_r = [\tau_r, \tau_{r+1})$ identifying the cell $C_r(\theta) \in \mathcal{P}_r$. These are history partitions, not necessarily partitions

generated by the latest observation alone. Observations do not depend on x, u, d ; active sensing and endogenous observation branching are excluded from the exact theorem (Section 9 gives a sound extension).

Policies and quantifiers. Policies are deterministic, measurable, nonanticipative, and adapted to the observation history; every such policy is represented exactly by signals $u_{r,C} \in L^\infty(I_r; U)$, $C \in \mathcal{P}_r$ — the realized control in mode θ being $u_\theta^\pi(t) = u_{r,C_r(\theta)}(t)$ on I_r . The robust quantifier order is

$$\exists \pi \in \Pi_{\mathcal{I}} \forall \theta \forall x_0 \in P_\theta \forall d \in L^\infty([0, T]; D_\theta) \forall t \in [0, T],$$

the adversary selecting mode, initial state, and measurable disturbance with knowledge of the announced policy; randomized policies are not included. Throughout, a belief is a nonempty set $B \subseteq \Theta \times \mathbb{R}^n$ of mode-state possibilities, not a probability distribution. The information-constrained viability kernel is $\mathcal{K}_{\mathcal{I}}^T = \{B : \exists \pi \forall \theta, x_0, d, t \ x_{\theta, x_0, d}^\pi(t) \in \mathcal{V}\}$, a set of beliefs; the audited initial state B_0 is viable exactly when $B_0 \in \mathcal{K}_{\mathcal{I}}^T$. Writing $\text{Lift}(K_{\text{full}}^T) = \{B \subseteq \Theta \times \mathbb{R}^n : \emptyset \neq B \subseteq K_{\text{full}}^T\}$ for the belief lifting of the full-information kernel (an ambient $\Theta \times \mathbb{R}^n$ -valued family), $\mathcal{K}_{\mathcal{I}}^T \subseteq \text{Lift}(K_{\text{full}}^T)$, and the inclusion is strict in general: the instance of Section 6 has every two-state subbelief viable — with the pairwise policies computed exactly there — while the three-state belief is not.

Shared symbols across the companion papers. The companion papers share this vocabulary with paper-local meanings; the letters below carry meanings here that differ there:

Here	In the companions
$\Gamma_{\mathcal{I}}(\lambda, a)$: the finite obstruction certificate margin (1)	the recourse certificate Γ_h (obstruction calculus; minimax duals); the alpha-vector witness sets Γ_k^Π (sufficiency)
λ_ℓ : the certificate weights	the Farkas multiplier and observer decay rate (obstruction calculus); the Farkas and dual weights (worked systems)
μ : the adversarial probability measure	Farkas multipliers (obstruction calculus); the adversarial measures of the minimax companion
K_{full}^T : the full-information viability kernel	the safe set and kernel K (obstruction calculus); the window depth and contact set $\mathcal{K}(B)$ (minimax duals); stock carrying capacities (forecast ladder)
ρ : the mesh relaxation parameter	the distributionally robust mixture parameter ρ (sufficiency; minimax duals); the net biomass multiplier (applied regime viability)
β_a : the stored per-label bound	the margin β (minimax duals)
τ : the observation time	the worst-case exit time τ^* (worked systems); the delay times τ (obstruction calculus)
σ : the kernel time variable	the floored exit count σ^* (worked systems); a noise scale (forecast ladder)

3. Adjoint safety rows and the safety value

Let $X_\theta(t, s)$ denote the state-transition matrix. For a label $a = (\theta, i, t) \in \mathcal{S} := \Theta \times \{1, \dots, s\} \times [0, T]$ define the input

kernel and margin

$$k_a(\sigma) = \mathbf{1}_{\{\sigma \leq t\}} B_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i,$$

$$\begin{aligned} \beta_a &= b_i - h_{P_\theta}(X_\theta(t, 0)^\top c_i) - \int_0^t c_i^\top X_\theta(t, \sigma) g_\theta(\sigma) d\sigma \\ &\quad - \int_0^t h_{D_\theta}(E_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i) d\sigma, \end{aligned}$$

where $h_Q(v) = \sup_{q \in Q} v^\top q$. For a policy π set $\mathcal{F}_a(\pi) = \int_0^T k_a(\sigma)^\top u_\theta^\pi(\sigma) d\sigma - \beta_a$.

Proposition 1 (robust rows). *π is robustly safe on $[0, T]$ if and only if $\mathcal{F}_a(\pi) \leq 0$ for every $a \in \mathcal{S}$.*

Proof. The identity is the linear solution formula: the initial-state and disturbance maximizations separate from the control term because the control signal in a mode does not depend on x_0 or d , so the rows are exact. For polytopic D_θ , a maximizing disturbance can be chosen measurably by selecting a maximizing vertex at each time, ties resolved by a fixed ordering. $\square$

Define the minimax safety violation $J_{\mathcal{I}}(T) = \min_{\pi \in \Pi_{\mathcal{I}}} \max_{a \in \mathcal{S}} \mathcal{F}_a(\pi)$.

Proposition 2 (value and nonviability). *The minimum exists: the policy-signal space is a finite product of weak-* compact sets $L^\infty(I_r; U)$ (each U a compact convex polytope, so the admissible signal sets are weak-* closed and bounded) and each $\mathcal{F}_a$ is weak-* continuous; the maximum over labels is attained on the compact label set $\mathcal{S}$ because $a \mapsto \mathcal{F}_a(\pi)$ is continuous in t under assumption (6) below. Consequently $B_0 \notin \mathcal{K}_{\mathcal{I}}^T$ if and only if $J_{\mathcal{I}}(T) > 0$.*

Proof. Existence: the policy-signal space is a finite product of weak-* compact sets $L^\infty(I_r; U)$ (each U a compact convex polytope, so the admissible signal sets are weak-* closed and bounded) and each $\mathcal{F}_a$ is weak-* continuous; the maximum over labels is attained on the compact label set $\mathcal{S}$ because $a \mapsto \mathcal{F}_a(\pi)$ is continuous in t under assumption (6) below. The equivalence is then the definition of the kernel: a violation-free policy exists exactly when the minimax value vanishes. The strict positive margin is a consequence of compactness and closed safety constraints, not an extra nonviability assumption. $\square$

4. The finite obstruction certificate

Define $\phi_U(g) := \min_{v \in U} g^\top v = -h_U(-g)$. Choose finitely many labels $a_\ell = (\theta_\ell, i_\ell, t_\ell)$, $\ell = 1, \dots, q$, and weights $\lambda_\ell \geq 0$, $\sum_\ell \lambda_\ell = 1$. For each active information cell $C \in \mathcal{P}_r$, pool the adjoint kernels $g_{r,C}(\sigma) = \sum_{\ell: \theta_\ell \in C} \lambda_\ell k_{a_\ell}(\sigma)$. The certificate margin is

$$\Gamma_{\mathcal{I}}(\lambda, a) = \sum_{r=0}^{R-1} \int_{I_r} \sum_{C \in \mathcal{P}_r} \phi_U(g_{r,C}(\sigma)) d\sigma - \sum_{\ell=1}^q \lambda_\ell \beta_{a_\ell}. \quad (1)$$

Remark 3 (semantics). If $\Gamma_{\mathcal{I}}(\lambda, a) > 0$, then no admissible information-adapted policy is safe: for every policy, $\sum_\ell \lambda_\ell \mathcal{F}_{a_\ell}(\pi) \geq \Gamma_{\mathcal{I}}(\lambda, a)$, so at least one stored label satisfies $\mathcal{F}_{a_\ell}(\pi) \geq \Gamma_{\mathcal{I}}(\lambda, a) > 0$ — an admissible realization violates facet i_ℓ at time t_ℓ . The violating label may depend on the policy; the finite list and its weights do not.

The certificate is not an adversarial tree: it is a finite, policy-independent set of adversarial tests combined by an adjoint identity. The pooled bound is tight as a minimum over policies: $\min_\pi \sum_\ell \lambda_\ell \mathcal{F}_{a_\ell}(\pi) = \Gamma_{\mathcal{I}}(\lambda, a)$, the pointwise minimizer being measurable by fixed-order vertex selection in the polytope U .

5. The continuous-to-finite theorem

The exact dual expression (1) is a genuine infinite-dimensional minimax statement, and we prove it in full rather than invoke it: the proof of (iv) below (complete version in the Supplementary Material) runs Sion's minimax theorem over the mixed extension to probability measures on the label set and transfers the result to finitely supported measures by weak-* density. The consequential result is a sound finite construction covering unrestricted measurable controls; classical infinite-dimensional duality (Anderson and Nash, 1987; Hettich and Kortanek, 1993) is the surrounding theory, not a load-bearing citation.

Outer control moments. Choose a control-time mesh $\mathcal{H} = \{J_1, \dots, J_H\}$ containing every sensing time. Introduce one vector $v_{j,C} \in U$ for every cell $C \in \mathcal{P}_{r(j)}$ active during J_j , where $r(j)$ is the unique observation interval containing J_j (the mesh contains every sensing time, so each block lies in one I_r), and let $Q = \sum_j |\mathcal{P}_{r(j)}|$ be the number of information-time blocks. For a label a define the finite row $A_{a,j,C} = \mathbf{1}_{\{\theta \in C\}} \int_{J_j} k_a(\sigma) d\sigma$, the block average $\bar{k}_{a,\mathcal{H}}(\sigma) = \frac{1}{|J_j|} \int_{J_j} k_a(s) ds$ on J_j , and the moment error

$$e_{a,\mathcal{H}} = \int_0^T h_U(\bar{k}_{a,\mathcal{H}}(\sigma) - k_a(\sigma)) d\sigma, \quad (2)$$

for which any verified upper bound $\bar{e}_a \geq e_{a,\mathcal{H}}$ may be used. For every measurable policy, its block averages belong to U , and

$$\sum_{j,C} A_{a,j,C}^\top v_{j,C} - \int_0^T k_a(\sigma)^\top u_{\theta(a)}^\pi(\sigma) d\sigma \leq \bar{e}_a. \quad (3)$$

Inequality (3) is the crucial orientation: finite feasibility is a necessary condition for continuous-time safety. The moments do not restrict the real controller to held inputs. For the sampled-data subclass whose policies are held constant on the blocks, the averaging term vanishes identically ($\int_{J_j} (k_a - \bar{k}_{a,\mathcal{H}}) d\sigma = 0$ on every block), so such a policy is continuously safe exactly when its block moments satisfy the unrelaxed rows: an exact reduction precisely when the row bounds are exact, $\bar{e}_a = 0$; with the error bounds (2) the program remains a relaxation even for held policies (in the instance the held witness policy attains the continuous value $\frac{3}{50}$ on every aligned mesh, while (5) returns $\frac{3}{50} - Th/4$).

Inner adversarial approximation. One may use a right-hand side $\beta_a^+ \geq \beta_a$ obtained from an explicit admissible initial state and disturbance signal rather than exact worst-case optimization. If $0 \leq \beta_a^+ - \beta_a \leq \delta_\beta$ uniformly over the labels used, the approximation is safe for nonviability: it weakens the adversary and enlarges the feasible controller set. For example, with $w_a(\sigma) = \mathbf{1}_{\{\sigma \leq t\}} E_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i$, on a disturbance mesh choose $d_{a,J} \in \arg \max_{d \in D_\theta} (\int_J w_a)^\top d$, yielding an explicit

piecewise-constant admissible disturbance; if D_θ has radius R_D about a fixed center and $\text{TV}(w_a) \leq V_D$, then

$$0 \leq \beta_a^+ - \beta_a \leq R_D V_D h_d, \quad (4)$$

the general budget being $\delta_\beta = \delta_{\text{init}} + \delta_{\text{dist}} + \delta_{\text{num}}$ for separately accounted initial-state and numerical error.

The finite program. Let $G \subset [0, T]$ be a finite time grid with $0, T \in G$ and every observation time in G ; the label set is the product $\mathcal{S}_G = \Theta \times \{1, \dots, s\} \times G$, with covering radius $h_t = \sup_{t \in [0, T]} \min_{\hat{t} \in G} |t - \hat{t}|$. Solve

$$\rho_{\mathcal{H}, G} = \min_{v, \rho} \rho \quad \text{s.t.} \quad A_a v - \beta_a^+ - \bar{e}_a \leq \rho \\ (a \in \mathcal{S}_G), v_{j, C} \in U. \quad (5)$$

The level ρ is an unrestricted real variable; negative values record strict safety slack.

Theorem 4 (continuous-to-finite). *Assume:*

- (H1) *uniformly over admissible policies, modes, and facets, $|\mathcal{F}_{\theta, i, t}(\pi) - \mathcal{F}_{\theta, i, t'}(\pi)| \leq L|t - t'|$ (6);*
- (H2) *the input kernels satisfy $\text{TV}(k_a) \leq V_U$ (7);*
- (H3) *the label maps $a \mapsto k_a$ and $a \mapsto \beta_a$ are continuous on the compact label set (k_a in L^1 norm, β_a absolutely; the L^1 form is the operative one — sup-norm continuity is neither needed nor true here, the step kernels below jumping at $\sigma = t$ for every t — while L^1 continuity is implied by bounded coefficients and enjoyed by every rational instance below);*
- (H4) *U has radius R_U about some fixed center;*
- (H5) *the time grid has covering radius at most h_t and the control mesh maximum length is h_u ;*
- (H6) *the adversarial approximation satisfies the bound above.*

Then the following value and sign statements hold under (H1)–(H6):

- (i) Certified value sandwich.

$$\rho_{\mathcal{H}, G} \leq J_{\mathcal{I}}(T) \leq \rho_{\mathcal{H}, G} + \max_{a \in \mathcal{S}_G} \bar{e}_a + \delta_\beta + L h_t. \quad (8)$$

Here δ_β must independently bound the full stored-label discrepancy in (H6), including any cell-label optimism; a cell position radius alone does not supply this bound. With (4) — the budget $\delta_\beta = \delta_{\text{init}} + \delta_{\text{dist}} + \delta_{\text{num}}$ — $J_{\mathcal{I}}(T) - \rho_{\mathcal{H}, G} \leq \max_a \bar{e}_a + R_D V_D h_d + \delta_{\text{init}} + \delta_{\text{num}} + L h_t$. Verified numerical row widths are added to the right side. The upper bound is realized by held controls, for which the averaging error vanishes identically, so the bound is charged by the error bounds the program actually used; the uniform choice $\bar{e}_a \equiv R_U V_U h_u$ — valid by (2) whenever $\text{TV}(k_a) \leq V_U$, since $\int_J |k_a - \bar{k}_a| \leq h_u \text{TV}(k_a|_J)$ and $h_U(g) \leq R_U \|g\|$ — recovers the announced $R_U V_U h_u$ form as a special case. The $\max \bar{e}_a$ form is the conservative one: a loose but verified choice inflates ρ 's relaxation and must inflate the charged bound with it (a counterexample is carried in Section 6).

- (ii) Sound finite certificates. If $\rho_{\mathcal{H}, G} > 0$, the program's dual supplies a finite certificate of nonviability of the original continuous-time system, covering every measurable information-adapted policy.

(iii) Completeness under refinement. If $B_0 \notin \mathcal{K}_{\mathcal{I}}^T$, then for every admissible refinement sequence along which h_u, h_d, h_t converge to zero and the stored-scenario widths $\delta_{\text{init}}, \delta_{\text{num}}$ do as well — near-maximizing stored initial states (Section 7) must be refined with the mesh for completeness to hold — the meshes eventually produce a positive finite certificate. Nonviability is thus semidecidable in this class for the input model in which the coefficients are exact rational or interval-enclosed piecewise-polynomial data with certified integration, total-variation, and uniform time-Lipschitz bounds: the semidecision procedure enumerates admissible meshes, solves each enclosed rational program, and halts on the first positive certificate — for instance, the dyadic schedule $h_u = h_d = h_t = T/2^k$ halts with certificate margin at least $\gamma/2$ once $2^{-k}(R_U V_U + R_D V_D + L)T < \gamma/2$ (with exactly stored scenarios, $\delta_{\text{init}} = \delta_{\text{num}} = 0$), where $\gamma = J_{\mathcal{I}}(T) > 0$ is the continuous value.

- (iv) Exact atomic dual representation.

$$J_{\mathcal{I}}(T) = \sup_{q < \infty, a_1, \dots, a_q, \lambda \in \Delta_q} \Gamma_{\mathcal{I}}(\lambda, a). \quad (9)$$

In particular, nonviability is equivalent to the existence of a positive finite atomic certificate.

- (v) Sparse witnesses. Let $r = \text{rank } A$ for the finite safety-row coefficient matrix of (5) — the A_a rows excluding the ρ -column and the actuator inequalities; the bound is linear-programming (LP) basic-solution theory (Chvátal, 1983; supplementary Step 6). Whenever $\rho_{\mathcal{H}, G} > 0$, a positive certificate exists using at most $r + 1 \leq mQ + 1$ safety labels, and it can retain any prescribed margin strictly below $\rho_{\mathcal{H}, G}$. If the lifted continuous-time kernels span a finite-dimensional space of dimension $r_{\mathcal{I}}$, a continuous-time obstruction admits one with at most $r_{\mathcal{I}} + 1$ labels. The relevant dimension is information–time rank, not m .

Proof (the complete steps are Steps 1–6 of the supplementary material, verified by the verification script). The lower bound holds because every admissible measurable policy's control bounds are outer-approximated by the block moments, so the moment program relaxes the controller's feasible set and its value cannot exceed the continuous safety value; the upper bound is realized by taking, on each block, the piecewise-constant control attained by the finite optimizer's block variables, whose exit from the continuous value is charged exactly by the program's own moment bounds $\bar{e}_a$ ($R_U V_U h_u$ under the uniform choice), the stored-scenario error δ_β , and the time-grid interpolation error $L h_t$. The asymmetry at its core is worth stating plainly: averaging arbitrary controls proves the lower bound; implementing piecewise-constant controls proves the upper bound. Confusing these two directions is what makes naïve infeasibility-by-discretization arguments unsound.

Proof of (iv) (exact atomic dual representation). The policy space $\Pi_{\mathcal{I}} = \prod_{r, C} \{u \in L^\infty(I_r; U)\}$ is weak-* compact and convex (Banach–Alaoglu: each factor is weak-* closed and bounded, U being a compact convex polytope), and the label set $\mathcal{S}$ is compact. Under the label-continuity hypothesis, $(a, \pi) \mapsto \mathcal{F}_a(\pi)$ is jointly continuous (π -continuity is the L^1 -pairing; a -continuity is uniform in π by the L^1 label continuity and boundedness of the U -values: $|\mathcal{F}_a(\pi) - \mathcal{F}_{a'}(\pi)| \leq M_U \|k_a - k_{a'}\|_{L^1} + |\beta_a - \beta_{a'}|$).

with $M_U := \sup_{v \in U} \|v\|$ (Hölder; the controls are M_U -bounded), the equicontinuity Arzelà–Ascoli consumes). Let $\mathcal{P}(\mathcal{S})$ carry the weak-* topology: by the Riesz representation it is the compact convex metrizable set of Borel probability measures. Since the integral of a continuous function over a probability measure lies within its min–max range and the Dirac measures attain both ends, $\max_a \mathcal{F}_a(\pi) = \max_\mu \int \mathcal{F}_a(\pi) d\mu$, and the bilinear $\int \mathcal{F}_a(\pi) d\mu$ — affine, hence simultaneously quasi-convex and quasi-concave, and continuous in each variable — meets the hypotheses of Sion’s minimax theorem, giving

$$J_{\mathcal{I}}(T) = \max_{\mu \in \mathcal{P}(\mathcal{S})} \min_{\pi} \int \mathcal{F}_a(\pi) d\mu,$$

the outer maximum attained. The value function $\mu \mapsto \min_{\pi} \int \mathcal{F}_a(\pi) d\mu$ is weak-* continuous: the family $\{a \mapsto \mathcal{F}_a(\pi)\}_{\pi}$ is uniformly bounded and uniformly equicontinuous, so by Arzelà–Ascoli weak-* convergence of measures makes the pairings converge uniformly over π . Finitely supported measures are weak-* dense in $\mathcal{P}(\mathcal{S})$, so this continuous value function has the same supremum over finitely supported measures as over all of $\mathcal{P}(\mathcal{S})$ — namely $J_{\mathcal{I}}(T)$. For $\mu = \sum_{\ell} \lambda_{\ell} \delta_{a_{\ell}}$ the inner minimum decomposes over information cells, and the cellwise infimum of an L^1 -pairing over U -valued signals equals the pointwise ϕ_U integral by measurable vertex selection (fixed vertex order in the polytope), which is exactly the representation $\Gamma_{\mathcal{I}}(\lambda, a)$ of (1). Nonviability $J_{\mathcal{I}}(T) > 0$ therefore yields a finitely supported μ with $\Gamma > J_{\mathcal{I}}(T)/2 > 0$; conversely $\Gamma > 0$ lower-bounds $J_{\mathcal{I}}(T)$ because $\max_a \geq$ the λ -weighted average. The complete seven-step verification of the remaining hypotheses is Step 5b of the Supplementary Material; the label-continuity input is the L^1 estimate above, verified in Section 6 for the audited kernels.

Corollary[complete dual calculus]. Under the hypotheses of Theorem 4: (a) every dual-feasible pair (λ, a) satisfies $\Gamma_{\mathcal{I}}(\lambda, a) \leq J_{\mathcal{I}}(T) \leq \max_a \mathcal{F}_a(\pi)$ for every policy — the lower bound is Remark 3, the upper is the weighted-average bound of the proof; (b) $J_{\mathcal{I}}(T) = \sup_{q, a, \lambda \in \Delta_q} \Gamma_{\mathcal{I}}(\lambda, a)$: the supremum over all measures is attained (Sion), and finitely supported measures approach its value arbitrarily finely — attainment by an atomic maximizer is an instance-level fact (in the instance of Section 6 three atoms attain it), not a general claim; (c) $B_0 \notin \mathcal{K}_{\mathcal{I}}^T$ if and only if some dual-feasible pair has $\Gamma_{\mathcal{I}}(\lambda, a) > 0$; every such pair obstructs every measurable information-adapted policy, and every obstruction arises this way. The duals therefore form the complete certificate system for the class, positivity being the only decision.

6. A three-branch delayed-observation instance carried exactly

Plant and exact delay threshold. Consider the planar double integrator $\dot{p} = v$, $\dot{v} = u$, with unit vectors $n_1 = (1, 0)$, $n_2 = (-\frac{3}{5}, \frac{4}{5})$, $n_3 = (-\frac{3}{5}, -\frac{4}{5})$, the hexagonal input set $U = \text{conv}\{\pm n_1, \pm n_2, \pm n_3\}$, equivalently $\{v : |v_2| \leq \frac{4}{5}, |2u_1 + u_2| \leq 2, |2u_1 - u_2| \leq 2\}$, safety $n_i^\top p \leq 2$ ($i = 1, 2, 3$) and $|v_1|, |v_2| \leq \frac{6}{5}$ — seven facets in all, $s = 7$ (three position, four velocity) — three possible initial states $p_j(0) = \frac{34}{25}n_j$, $v_j(0) = n_j$, no disturbance, and the mode unobserved until time τ , when it is revealed exactly.

If j is known, $u = -n_j$ for one second and then $u = 0$ keeps the maximum critical position at $\frac{34}{25} + \frac{1}{2} = \frac{93}{50} = 1.86 < 2$: all three physical states are indefinitely viable under full information (throughout the braking leg $\|v(t)\|_{\infty} \leq 1 \leq \frac{6}{5}$; the noncritical facets keep $n_i^\top p < 0$ along the n_j -ray; $-n_j \in U$). Take three witness labels, each using its own position facet at $t_* = \tau + 1$, with kernels $k_j(s) = \mathbf{1}_{\{s \leq t_*\}}(t_* - s)n_j$ and $\beta_j = \frac{16}{25} - t_*$, weighted by $\lambda_1 = \frac{3}{8}$, $\lambda_2 = \lambda_3 = \frac{5}{16}$, which satisfy $\sum_j \lambda_j = 1$ and $\sum_j \lambda_j n_j = 0$. Before observation the pooled kernel is identically zero; after observation each mode has its own control and $\phi_U(\lambda_j(t_* - s)n_j) = -\lambda_j(t_* - s)$. Hence

$$\Gamma(\tau) = - \int_{\tau}^{\tau+1} (\tau+1-s) ds - \left(\frac{16}{25} - \tau - 1\right) = \tau - \frac{7}{50}, \quad (10)$$

so $\tau > \frac{7}{50}$ implies $B_0 \notin \mathcal{K}_{\mathcal{I}}$. Conversely, $u = 0$ before observation and $u = -n_j$ for one second afterward keeps the maximum critical position at $\frac{34}{25} + \tau + \frac{1}{2}$, safe exactly when $\tau \leq 0.14$. Thus $\tau_{\max} = \frac{7}{50} = 0.14$ is exact (equality viable: the safety set is closed), and for horizons $T \geq \tau + 1$ the upper-bound policy — full braking from τ on, stopping at $\tau + 1$ from $p = (\frac{34}{25} + \tau)n_j$ — shows every row value at most $\tau - \frac{7}{50}$, so $J_{\mathcal{I}}(T) = \tau - \frac{7}{50}$.

Pairs are viable; the blind-window exit test misses. At $\tau = \frac{1}{5}$, the common control $u = 0$ keeps all three branches safe throughout the blind window: $n_j^\top p_j(0.2) = \frac{39}{25} = 1.56 < 2$. The pair claims are carried exactly. For $\{1, 2\}$ (and $\{1, 3\}$ by the reflection $v_2 \mapsto -v_2$): brake *through* the blind window with the common control $u = -(n_1 + n_2) = (-\frac{2}{5}, -\frac{4}{5}) \in U$ (both branches decelerate along their normals at $\frac{2}{5}$), then brake each branch along its own normal, $u = -n_j$ from τ to n_j -rest — the critical position peaks at $\frac{12345}{6250} = 1.9752 < 2$ on both branches, slack $\frac{31}{1250}$. For $\{2, 3\}$: hold $u = n_1$ through the blind window (both branches decelerate at $\frac{3}{5}$), then brake each branch along its own normal, $u = -n_j$ from τ — the critical position peaks at $\frac{2419}{1250} = 1.9352$, slack $\frac{81}{1250}$. (Two tempting variants fail: holding $u = 0$ through the blind window in $\{1, 2\}$ peaks at $\frac{2719}{1250} = 2.175 > 2$ with the printed mid-brake, and at $\frac{103}{50}$ even with full braking afterward; inserting a $-\frac{3}{5}n_j$ leg after τ in $\{2, 3\}$ peaks at $\frac{2501}{1250} = 2.0008 > 2$ at $t = \frac{29}{25}$.) All seven facets hold at every time on every in-belief branch of every pair policy on $[0, \frac{6}{5}]$: every velocity value along $\{1, 2\}$ satisfies $\|v\|_{\infty} \leq 1$ and along $\{2, 3\}$ $\|v\|_{\infty} \leq \frac{4}{5}$, both strict against $\frac{6}{5}$, and the worst slacks are the peak rows $\frac{31}{1250}$ and $\frac{81}{1250}$; braking to n_j -rest leaves a harmless transverse velocity of $\frac{4}{25}$ (its drift only decreases the non-critical coordinates). The blind control $u = n_1$ does drive the off-belief branch 1 to $|v_1| = \frac{6}{5}$ exactly at τ — irrelevant to the pair claim (branch 1 is not in the belief), and indeed that control could not keep branch 1 safe: its peak would be $\frac{23}{10} = 2.3 > 2$. No exit occurs before information arrives; nevertheless $\Gamma(0.2) = 0.06 > 0$ — the best post-observation recourse cannot recover all branches. The obstruction is not that information arrives after unavoidable physical exit; it is that information arrives after some indistinguishable branch has necessarily accumulated more braking liability than the remaining control authority can discharge.

The finite program and its exact witnesses. Set $\tau = \frac{1}{5}$, $T = \frac{6}{5}$, $h = \frac{1}{10}$: two common pre-observation control blocks and ten post-observation blocks for each of three modes, so $Q = 32$, $mQ = 64$. For the endpoint posi-

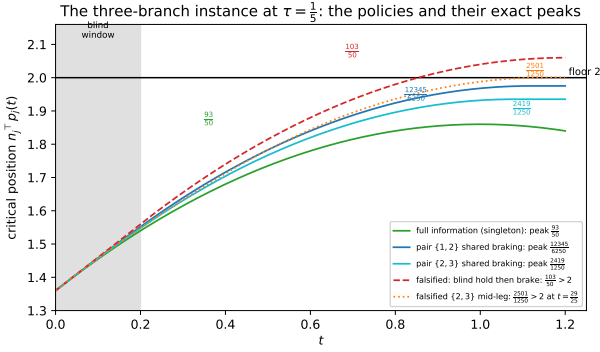


Figure 1: The audited policies on the critical coordinate $n_j^\top p_j(t)$ at $\tau = \frac{1}{5}$, every peak the exact rational of the text: the full-information singleton peaks at $\frac{93}{50}$; the pair policies peak below the floor ($\frac{12345}{6250}$, $\frac{2419}{1250}$, slacks $\frac{31}{1250}$ and $\frac{81}{1250}$); the two falsified variants cross it ($\frac{103}{50}$, $\frac{2501}{1250}$ at $t = \frac{29}{25}$). Generated by the assertion script `paper2_comp_certification_figures_v1.py`.

tion rows the exact moment error is $e = 12 \cdot h^2/4 = \frac{3}{100}$ (the kernel is affine on each of the $T/h = 12$ mesh intervals and $h_U(\alpha n_j) = |\alpha|$); the endpoint right-hand side is $\beta = \frac{16}{25} - \frac{6}{5} = -\frac{14}{25}$, relaxed to $\beta + e = -\frac{53}{100}$. The exact continuous certificate has margin $\frac{3}{50} = 0.06$; the finite moment certificate has margin $\frac{3}{100} = 0.03$.

The dual witness is exact. Order the input facets by $F = [(0, 1); (0, -1); (2, 1); (2, -1); (-2, 1); (-2, -1)]$, $f = (\frac{4}{5}, \frac{4}{5}, 2, 2, 2, 2)$, and define $\eta_1 = (0, 0, 0, 0, \frac{1}{4}, \frac{1}{4})$, $\eta_2 = (0, \frac{1}{2}, 0, \frac{3}{10}, 0, 0)$, $\eta_3 = (\frac{1}{2}, 0, \frac{3}{10}, 0, 0, 0)$, satisfying $F^\top \eta_j = -n_j$ and $f^\top \eta_j = 1$. For a post-observation interval J_k put $a_k = \int_{J_k} (T-s) ds$ and $\mu_{j,k} = \lambda_j a_k \eta_j$. Before observation, stationarity holds because $\sum_j \lambda_j n_j = 0$; after observation, because $\lambda_j a_k n_j + F^\top \mu_{j,k} = 0$. Since $\sum_k a_k = \frac{1}{2}$, the Farkas contradiction is $\lambda^\top (\beta + e) + \mu^\top f_{\text{blk}} = -\frac{53}{100} + \frac{1}{2} = -\frac{3}{100} < 0$ — an exact rational certificate. The primal witness is equally exact: hold the pre-observation blocks at 0 and the post-observation blocks of mode j at $-n_j$; the endpoint row then reads $-\frac{1}{2} + \frac{14}{25} - \frac{3}{100} = \frac{3}{100}$, and every other row lies at or below the same value. Three structural facts make this transparent: the velocity and non-critical position rows are continuously safe at the witness ($\|v_\theta(t)\|_\infty \leq 1$ against the $\frac{6}{5}$ boundary; $n_i^\top n_j < 0$ for $i \neq j$ with the witness trajectory on the n_j ray), so their row values $\mathcal{F}_a(\hat{\pi}) - \bar{e}_a$ are nonpositive; the endpoint critical rows attain $\rho = \frac{3}{100}$ with equality; and the remaining critical rows interpolate monotonically between the two (the braking profile $\zeta(t) = \frac{39}{25} + (t - \tau) - \frac{(t - \tau)^2}{2}$ increases to its endpoint maximum) — these are verified row-by-row, with the campaign script checking all 465 rows exactly at this mesh and archiving the slacks (Section 14).

Instance arithmetic, collected. The exact rational chain of this subsection is: the pooling weights $\lambda = (\frac{3}{8}, \frac{5}{16}, \frac{5}{16})$ with $\sum_j \lambda_j = 1$ and $\sum_j \lambda_j n_j = 0$; the witness margins $\beta_j = \frac{16}{25} - t_*$ and the value $\Gamma(\tau) = \tau - \frac{7}{50}$ in (10), so $\tau_{\max} = \frac{7}{50}$ with equality viable; the moment error $e = Th/4$, giving at $h = \frac{1}{10}$ the relaxed right-hand side $\beta + e = -\frac{53}{100}$, the dual Farkas value $-\frac{53}{100} + \frac{1}{2} = -\frac{3}{100} < 0$, and the primal endpoint row $-\frac{1}{2} + \frac{14}{25} - \frac{3}{100} = +\frac{3}{100}$; and the law $\rho(h) = \frac{3}{50} - Th/4$ on every uniform aligned mesh (the level unrestricted; negative on coarser non-uniform aligned meshes). Every identity is an exact rational statement, machine-verified in Section 14.

Solver-assisted, exactly certified mesh study. For each mesh the program (5) is built at its claimed dimensions and solved by HiGHS; optimality is then certified in exact rational arithmetic by the witness pair above (primal feasibility of every row; dual stationarity and dual objective $-\rho(h)$), so the duality gap closes exactly and the solver’s returned value is a computational confirmation of a proved optimum, not an observation. Table 1 reports the campaign: exact values $\rho(h) = \frac{3}{50} - Th/4$ (nonnegative on this family), solver values, iteration counts, and wall-clock times as measured on the verification machine.

On the audited family of uniform meshes $h \in \{\frac{1}{5}, \frac{1}{10}, \frac{1}{20}, \frac{1}{50}, \frac{1}{100}\}$ (each dividing both τ and T), the values obey the exact law $\rho(h) = \frac{3}{50} - Th/4$ for $h < \frac{1}{5}$ and vanish at $h = \frac{1}{5}$: the whole refinement gap is the control-moment error $Th/4$ of the endpoint rows: the finite value lies below the continuous margin by exactly that error and converges to it from below as $h \downarrow 0$ ($\rho \leq J$ always — the program is a relaxation; Table 1 itself shows $0.03 < 0.06$). The $\max\{0, \cdot\}$ in the table’s law is inactive on this family: for $h \leq \frac{1}{5}$ the raw value $3/50 - Th/4$ is already nonnegative, vanishing at $h = \frac{1}{5}$; the level is unrestricted, and coarser non-uniform meshes record negative values (below). For the record, the instance’s hypothesis constants are $R_U = 1$ (the hexagon’s circumradius), $V_U = \tau + 1 = \frac{6}{5}$ (the endpoint kernels are unit-slope ramps of height t^*), $\delta_\beta = 0$ (β is computed exactly; no stored-scenario error), and $L = T + 1 = \frac{11}{5}$ — the kernel difference $k_t - k_{t'}$ has L^1 norm at most $T|t - t'|$ (it is *not* supported on $[t', t]$: the position rows also differ by $(t - t')n_j$ on $[0, t']$; the integral bound is what (6) consumes), while $|\beta_t - \beta_{t'}| = |t - t'|$; the sandwich (8) thus bounds $J \leq \rho(h) + Th/4 + \frac{11}{5}h_t$, while the witness pair proves the realized gap is exactly $Th/4$. The $\max \bar{e}_a$ form of (8) is load-bearing: with the loose but verified choice $\bar{e} = e + \frac{1}{2}$ the program returns $\rho = -\frac{47}{100}$ at $h = \frac{1}{10}$, and the certified bound $J \leq \rho + \bar{e} + \delta_\beta + Lh_t = \frac{17}{100} \geq \frac{3}{50}$ stays valid, while the uniform special case $R_U V_U h_u$ would claim $J \leq -\frac{6}{25}$ — false at $J = \frac{3}{50}$. The domain is larger than the campaign: for every uniform aligned mesh (h dividing both τ and $T - \tau$) the same primal–dual witness pair proves $\rho(h) = \frac{3}{50} - Th/4$ — the dual objective $\sum_k a_k$ telescopes to $(T - \tau)^2/2$ independently of h — confirmed exactly at the sixth aligned mesh $h = \frac{1}{15}$, where $\rho = \frac{1}{25}$ ($Q = 48, 687$ inequalities). No floor is claimed off the uniform family: on an aligned mesh with unequal blocks the endpoint moment error is $\sum_J |J|^2/4$, and the allowed two-block mesh $[0, \frac{1}{5}], [\frac{1}{5}, \frac{6}{5}]$ gives $\rho = \frac{3}{50} - (\frac{1}{100} + \frac{1}{4}) = -\frac{1}{5}$, certified by the same witness pair (primal row $-\frac{1}{2} + \frac{14}{25} - \frac{26}{100}$; dual $\frac{1}{2} + (\beta + \frac{26}{100}) = \frac{1}{5} > 0$). For affine kernels the block moment error is exactly $|J|^2/4$ per block, so on this instance $e = \sum_J |J|^2/4$ on *every* mesh, uniform or not, and equal blocks minimize $\sum_J |J|^2$ at fixed count: time-refinement here has no conflict structure to target. The conflict-interval decomposition (12) is a different axis — it prices *information* cell splits, not control-block splits — and it is the axis the ladder subsection below reads prescriptively. All dimension counts ($mQ + 1$ variables; $Ms(N_t + 1) + p_U Q$ inequalities), all witness inequalities, and the closed-form law are machine-verified in exact rational arithmetic by the verification scripts of Section 14, which regenerate Table 1 verbatim and re-run the solver campaign.

h	blocks Q	variables $mQ+1$	rows $Ms(N_t+1) + p_U Q$	exact $\rho(h)$	solver ρ	iterations	time (s)
0.20	16	33	243	0	0.000000000	36	0.024
0.10	32	65	465	$\frac{3}{190} = 0.030$	0.030000000	78	0.014
0.05	64	129	909	$\frac{200}{200} = 0.045$	0.045000000	141	0.043
0.02	160	321	2241	$\frac{27}{590} = 0.054$	0.054000000	501	0.252
0.01	320	641	4461	$\frac{57}{1000} = 0.057$	0.057000000	1062	1.015

Table 1: Solver-assisted, exactly certified mesh study on the three-branch instance ($\tau = \frac{1}{5}$, $T = \frac{6}{5}$): program dimensions, exact optimal values, returned values (agreement to at most 4.2×10^{-17}), simplex iterations, and wall-clock time (single runs, scipy HiGHS dual simplex, on the verification machine; iteration counts and times may vary with solver version and settings; “rows” counts LP inequalities — safety plus actuator, with $s = 7$, $p_U = 6$, $N_t = T/h$). Optimality of each exact value is certified by the exact primal–dual witness pair, independent of the solver. At $h = 0.2$ the witness is an exact optimality witness of value zero; at the four finer meshes it is a positive obstruction certificate. The exact continuous value is 0.06; the three-label support of each positive certificate is an instance-specific fact of this symmetric campaign, distinct from the general $r + 1$ bound.

7. Computation, complexity, and independent verification

With Q information–time blocks and $N_t + 1$ time labels, the program has $mQ + 1$ variables and at most $Ms(N_t+1) + p_U Q$ inequalities (p_U input-polytope facets); it is polynomial-time in these dimensions and the encoded rational data size (weakly polynomial bit complexity) for the exact rational benchmark programs, and for interval-enclosed inputs once the enclosures are fixed — conditional on the validated coefficient bounds and certified row integrals. Initial-state dependence is affine, so no vertex enumeration of P_θ is required: at each label solve $\max_{\xi \in P_\theta} c_i^\top X_\theta(t, 0)\xi$ and store a maximizing or near-maximizing point — the verifier needs only membership of the stored point in P_θ , not global optimality; near-optimality bears on the convergence guarantee, not on a positive certificate’s validity. The same holds for stored disturbance signals.

For rational outer rows $\hat{A}v \leq \hat{b}$, $F_{\text{blk}}v \leq f_{\text{blk}}$, a residual-bounded certificate test suffices: with $\lambda, \mu \geq 0$, $r = \hat{A}^\top \lambda + F_{\text{blk}}^\top \mu$, $c = \lambda^\top \hat{b} + \mu^\top f_{\text{blk}}$, the condition

$$c + \sum_j h_U(-r_j) < 0 \quad (11)$$

is sufficient, since any feasible v satisfies $-\sum_j h_U(-r_j) \leq r^\top v \leq c$: the upper bound is the multiplier combination, the lower bound is termwise minimization over U , and the two displayed inequalities make the sign convention auditable. The actuator rows $F_{\text{blk}}v \leq f_{\text{blk}}$ already represent $v_j \in U$; their multipliers in the residual test are redundant but legitimate, and the verifier uses both paths as independent cross-checks. All integration, rounding, and model-enclosure errors must be included before applying the test, and uncertain row data must be relaxed toward a larger feasible controller set: overestimating the worst disturbance contribution can create a false obstruction — the sign convention of the $\beta_a^+ \geq \beta_a$ relaxation of Section 5: a nonmaximizing stored disturbance gives a larger β_a^+ , hence a weaker constraint, which is the sound direction for a lower bound on $J_{\mathcal{I}}(T)$. A deletion-based sparsification attempts supported labels for deletion one at a time in a fixed order, keeping a deletion only if the prescribed target margin survives the feasibility re-solve; it terminates with an irredundant (not claimed minimal) supported label set, at one feasibility solve per safety row; exact rank compression requires verified rank, and floating-point singular-value truncation must be charged

to the error budget. If $J_{\mathcal{I}}(T) = \gamma > 0$, a mesh with $\max_a \bar{e}_a + R_D V_D h_d + \delta_{\text{init}} + \delta_{\text{num}} + L h_t < \gamma$ guarantees detection — an explicit margin-dependent resolution bound (mesh resolution as a function of the margin γ), not a certificate-size bound in general; the uniform choice $\bar{e}_a \equiv R_U V_U h_u$ gives the announced special case.

8. Information–time rank: input dimension does not bound the witnesses

Proposition 5 (arbitrarily large minimal obstructions with one input). *For every integer $q \geq 1$ there is a continuous-time system with scalar input $u(t) \in [0, 1]$ and $q + 1$ indistinguishable modes such that every proper sub-belief is viable, the full $(q+1)$ -mode belief is not, and every obstruction certificate must involve all $q + 1$ modes.*

Proof. Let $T = q$, $I_j = [j - 1, j]$. Mode $j \leq q$ carries the scalar state x obeying $\dot{x} = \mathbf{1}_{I_j}(t)(u - 1)$, $x(0) = 0$; mode $q + 1$ carries $\dot{x} = -u$, $x(0) = q - \varepsilon$, $0 < \varepsilon < 1$. Require $0 \leq x \leq q$ — the terminal bounds are the binding ones, each x being monotone along its mode — and there are no observations. Writing $w_j = \int_{I_j} u$, safety requires $-w_j \leq -1$ for $j \leq q$ and $\sum_j w_j \leq q - \varepsilon$; summing gives $0 \leq -\varepsilon$, a contradiction. If mode $q + 1$ is removed, $u \equiv 1$ is safe; if mode $j \leq q$ is removed, $u = 0$ on I_j and $u = 1$ elsewhere is safe. A certificate supported on fewer than $q + 1$ modes would also certify the corresponding proper subbelief, contradicting its viability, so every obstruction certificate involves all $q + 1$ modes. Equal weights give certificate margin $\varepsilon/(q + 1)$; the kernel rank is q , and $q + 1$ witnesses are necessary. $\square$

A statement such as “at most $m + 1$ compatible states suffice to witness nonviability” (Helly’s theorem, Helly, 1923) is valid for convex common-action sets in $\mathbb{R}^m$ but false for a common blind control function: the correct dimension counts independent temporal and informational control decisions. This distinction is consequential for review intervals, delayed sensing, and certificate-size claims; its discrete counterpart is a scoping caveat of the finite audits of Abaee (2026, Worked systems for the obstruction calculus).

9. Refinement erosion and oracle recourse

Suppose a coarse information cell C is split into finer cells $C_1, \dots, C_k$. Because $\phi_U(g_1 + \dots + g_k) \geq \sum_j \phi_U(g_j)$, the

same witness satisfies $\Gamma_{\text{coarse}} \geq \Gamma_{\text{fine}}$, with the exact difference

$$\int \left[\phi_U \left(\sum_j g_{C_j} \right) - \sum_j \phi_U(g_{C_j}) \right] dt. \quad (12)$$

For $U = [-\bar{u}, \bar{u}]^m$ this is $\bar{u} \int [\sum_j \|g_{C_j}(t)\|_1 - \|\sum_j g_{C_j}(t)\|_1] dt$: the certificate identifies the precise time intervals on which indistinguishable branches demand conflicting control effort. Equation (12) is the change in the same witness's margin under refinement, not a claim about the optimized viability value: a coarse cell can strengthen an obstruction, and splitting it can dissolve the certificate while the system's verdict is unchanged. The inequality is proved above by pointwise minimization; on rational instances it is additionally exactly checkable, and the verification record confirms $\phi_U(g_1 + g_2) \geq \phi_U(g_1) + \phi_U(g_2)$ on 3,240 deterministic rational pairs for the hexagonal U of Section 6 as a software regression test, not as evidence for the identity. The discrete counterpart of this axis is the master monotonicity of the worked-systems companion (Abaee, 2026, Worked systems for the obstruction calculus): its three axes (observations, policies, memory) enlarge kernels exactly as refinement here tightens the sandwich, and positivity is non-monotone on both sides.

The exact theorem assumes exogenous mode observations. A useful sound extension does not: if finitely many admissible initial-state/disturbance scenarios are provably indistinguishable until τ , give the controller after τ separate arbitrary control signals for every stored scenario — an oracle relaxation of the true policy class. Scenarios are stored initial-state/disturbance pairs, indistinguishable through τ in the sense that they generate identical observation histories; after τ the oracle reveals the realized scenario, and separate controls are granted even where scenarios remain observationally identical. The soundness direction: every real output-feedback policy induces one common pre- τ control, the oracle class contains every such policy, and nonviability in the enlarged class transfers down. If

$$\int_0^\tau \phi_U \left(\sum_\ell \lambda_\ell k_\ell \right) dt + \sum_\ell \int_\tau^T \phi_U(\lambda_\ell k_\ell) dt - \sum_\ell \lambda_\ell \beta_\ell > 0, \quad (13)$$

with β_ℓ computed from the explicitly stored scenario, then no actual output-feedback policy can be safe. This covers arbitrary sensing after a known blackout; it is sound but generally incomplete.

A third extension replaces exact scenario storage — and the exact revelation of the audited model — by a certified *belief cell*: a set $C \subseteq \Theta \times \mathcal{X}$, carried with a polytope enclosure, that the observation history certifies contains the true mode-state pair. — a set-valued observer with a guaranteed error radius, in the sense of the bounding approaches to estimation (Milanese et al., 1996).

Proposition 6 (conditional belief-cell certificates). *Let each cell label be an explicitly verified outer row enclosure for its compatible mode-state pairs. With the original bridge's (H6) checked for these labels, including a separate bound on the full label discrepancy, the corresponding finite program is linear (or conic for certified cell support functions) and the lower bound and positive-margin obstruction of Theorem 4 apply to the cell-filtration policy class. An upper sandwich requires the same separately verified*

(H6) budget and an implementable held-policy comparison; it does not follow from a cell radius alone. Splitting cells can change both the row labels and the control-kernel grouping, so (12) accounts for the latter term only. A safety-margin claim for a primal witness requires its actual held-policy row slack and an enclosure error no greater than that slack; a dual margin survives only after all enclosure errors charged by its normalized weights remain smaller than the margin.

Proof. Every real compatible pair obeys a valid one-sided outer row bound, so its policy supplies a feasible relaxation; a positive relaxed optimum obstructs the actual cell-adapted policy class. To turn the relaxation into an upper value bound, apply the bridge's held-policy comparison with an independently checked (H6) discrepancy. Splitting a cell also replaces its label by subcell labels, independently of any control-kernel subadditivity. For example, with $\dot{x} = 0$, $x \in C = [0, 1/2]$, upper facet $x \leq 1$, and $\beta_x = 1 - x$, the label $\sup_C \beta_x + 1/4 = 5/4$ becomes $1/2$ on the singleton $\{1/2\}$ even though the control kernel is zero. This changes the label term by $3/4$ while the kernel term in (12) stays zero. The margin conditions follow by the row-wise enclosure inequality and nonnegative normalized dual weights, not by an unverified radius identification. $\square$

On the instance the belief cells are the compatible-mode sets, and the audited delay hierarchy is exactly the cell-refinement family: the full cell $\{1, 2, 3\}$ obstructs past $\frac{7}{50}$, the two-mode cells $\{1, 2\}$ and $\{2, 3\}$ are viable to $\frac{15 - \sqrt{183}}{6}$ and $\frac{10 - \sqrt{58}}{6}$, and the singleton cells never obstruct (Proposition 7). The audited instance has pair-policy row slacks $31/1250$ and $81/1250$ at $\tau = 1/5$ and a triple dual margin $3/50$. For a proposed noisy state cell these numbers can be used as budgets only after its actual held-policy slack and full row-label errors are enclosed separately; a nominal position radius is not itself a proof of the needed error bound. The companion deadline threshold concerns its stated exact-state schedule, not arbitrary noisy state cells.

10. Reading the certificate prescriptively: the shared-authority ladder

A positive certificate is not only a verdict; its weights, kernels, and facet multipliers encode what a controller would have to change. On the instance the decoding is exact end to end.

Proposition 7 (shared-authority ladder). (i) *The pooling weights satisfy $\sum_j \lambda_j n_j = 0$ with $\lambda_j > 0$; hence for every common control $w \in U$ some branch has $n_j^\top w \geq 0$: no common control brakes all three branches, and the pooled blind-window kernel — the certificate's zero charge on $[0, \tau]$ — is exactly this unbrakeability.* (ii) *The optimal shared braking authorities $t^*(B) = \max_{w \in U} \min_{j \in B} (-n_j^\top w)$ are $t^*(\{1, 2\}) = t^*(\{1, 3\}) = \frac{2}{5}$, attained at $u = (-\frac{2}{5}, -\frac{4}{5})$ and its v_2 -reflection, and $t^*(\{2, 3\}) = \frac{3}{5}$, attained at $u = n_1$: the blind-window controls of the pair policies of Section 6 are exactly these optima.* (iii) *Braking at authority t^* through the blind window and $u = -n_j$ afterward holds the critical-position peak at $\frac{93}{50} + (1 - t^*)\tau + \frac{(t^*)^2 - t^*}{2}\tau^2$, increasing in τ on $[0, 1/t^*]$; the pair delay thresholds — where the*

peak reaches 2 — are $\tau_{12} = \tau_{13} = \frac{15-\sqrt{183}}{6} \approx 0.245$ and $\tau_{23} = \frac{10-\sqrt{58}}{6} \approx 0.397$, against the triple's $\tau_{\max} = \frac{7}{50}$, with singletons unbounded (the full-information peak $\frac{93}{50}$ is delay-independent). For every τ at or below its threshold the critical position stays ≤ 2 with the velocity rows strict; the verification script verifies the full seven-facet scan exactly at rational surrogates below each threshold ($\tau = \frac{6}{25}$ and $\tau = \frac{39}{100}$).

Proof. (i) is the pooling identity itself: $\sum_j \lambda_j (n_j^\top w) = (\sum_j \lambda_j n_j)^\top w = 0$ cannot have every summand strictly negative. (ii) The objective $\max(n_i^\top w, n_j^\top w)$ is piecewise linear, so its minimum over the hexagon is attained at a vertex or on the equal-projection segment $(n_i - n_j)^\top w = 0$. On that segment both authorities coincide: $w = -c(1, 2)$ for $\{1, 2\}$ (the facet $|u_2| \leq \frac{4}{5}$ binds at $c = \frac{2}{5}$) and its mirror for $\{1, 3\}$; $w = (c, 0)$ for $\{2, 3\}$ (the vertex n_1 binds at $c = 1$); the vertices themselves give at most $\frac{7}{25}$ for the mixed pairs and $\frac{3}{5}$ for $\{2, 3\}$. (iii) Along the optimal schedule the window exit state is $\frac{34}{25} + \tau - t^* \tau^2/2$ at velocity $1 - t^* \tau$ on the critical ray, and full braking adds $(1 - t^* \tau)^2/2$; collecting terms gives the displayed quadratic, whose τ -vertex sits at $1/t^* \geq 1$, so it increases on the audited range, and the thresholds are its exact roots in $\mathbb{Q}(\sqrt{183})$ and $\mathbb{Q}(\sqrt{58})$ ($6\tau^2 - 30\tau + 7 = 0$ and $6\tau^2 - 20\tau + 7 = 0$). The facet scans are exact rational evaluations at the stated candidates. $\square$

The prescriptive reading follows directly. The obstruction is removed by shortening the observation window below the relevant rung (the pairs already tolerate $\tau = \frac{1}{5}$ under shared braking alone — the audited delay is strictly inside both pair rungs), by adding observers (each information refinement deletes rungs, and the conflict-interval decomposition (12) prices exactly which splits pay), or by reallocating authority where the certificate charges it (the post-window controls already brake at the hexagon's face $-n_j \in U$, so the binding resource is the shared window, not the post-observation actuator). The dual witness agrees: the η_j place their mass on the braking facets $\mp n_j$, and the primal witness brakes every mode to rest — the certificate is the blind window's unbrakeability, propagated to the deadline. The decode is quantitative: each redesign lever carries an exact rate.

Proposition 8 (what the printed redesign witness establishes). *On the audited $a = 1$ instance the delay threshold is $\tau_{\max} = 7/50$, with a separate viable equality witness. The original dual's pooled pre-window kernel vanishes because $\sum_j \lambda_j n_j = 0$; thus its pre-window dual expression is insensitive to a shared-control-set enlargement. Its velocity-row multipliers are zero. These are properties of that certificate, not an exact viability sensitivity law for arbitrary redesigns. For $a > 1$, the extrapolated expression $\Gamma_a(\tau) = \tau - 7/50 - (a-1)/2$ and its zero locus are only algebraic plots of the selected expression, not an optimized viability threshold or a valid finite-horizon certificate on all plotted rungs. In particular, the equality-viable assertion for $a = 2, \tau = 16/25$ is false at the fixed horizon $T = 6/5$.*

Proof. The $a = 1$ identities and primal equality policy are the original instance's audited witnesses. For the dilated counterexample, let $\lambda = (3/8, 5/16, 5/16)$; the weighted

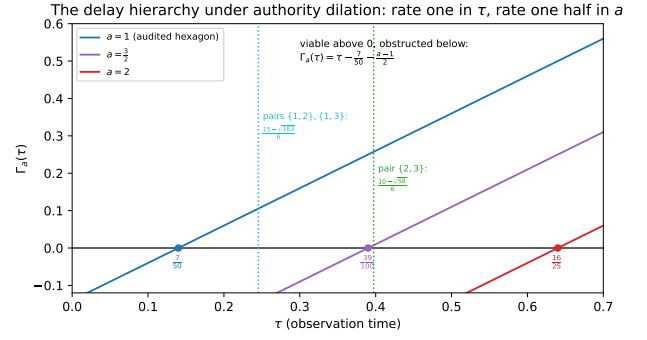


Figure 2: Zeros of the selected algebraic dilation expression, not certified viability thresholds under authority dilation: $\Gamma_a(\tau) = \tau - \frac{7}{50} - \frac{a-1}{2}$ crosses zero at $\frac{7}{50}, \frac{39}{100}, \frac{16}{25}$ for $a = 1, \frac{3}{2}, 2$ (formal rates only; the $a > 1$ equality points are not viable under the stated full horizon); the pair delay thresholds $\frac{15-\sqrt{183}}{6}$ and $\frac{10-\sqrt{58}}{6}$ mark the shared-braking rungs of Proposition 7. The original assertion script checks the plotted expression, not these dilated systems' optimized thresholds.

critical position is $34/25 + t$ throughout every common-control blind window because the weighted normals cancel. At $\tau = 16/25$ it equals the safety ceiling 2. For $s = 1/10$ after revelation, even granting each branch independent control in $2U$, its weighted position is at least $34/25 + 37/50 - 2(1/10)^2/2 = 209/100 > 2$, at time $37/50 < T$. Thus some branch exits before the declared horizon. The former proof's $\max_{u \in aU} \langle u, n_j \rangle = -a$ is false (the *minimum* is $-a$), and scaling both sides of $Fu \leq f$ to $aFu \leq af$ leaves U unchanged; a representation of aU instead uses $Fu \leq af$. $\square$

These formal expression rates must not be interpreted as exact redesign effects on the optimized viability boundary.

11. The reference library

The finite side of the pipeline is packaged as **viacert**, a standard-library-only Python package (no third-party dependencies) using exact integer and rational arithmetic. Its design follows three principles. *Exactness*: the finite audits and exact certificate verifiers use integer and rational arithmetic; floating-point solver output is never used as a proof object. *Fidelity to the verified recursion*: the finite-horizon selector is the calculus's backward recursion (Abaee, 2026, An obstruction calculus) with the no-repeat protocol threaded as a parameter, matching the audited implementations rather than a reformulation. *Pluggability*: new systems enter through a **FiniteSystem** interface (states, actions, safe set, transition, null action); the audited systems — the shared two-coordinate audit system, the continuous benchmark caps, the certainty-equivalence law pair, and the review grid — ship as default instances.

The public surface comprises the recursion module (viability verdicts, kernels, and step-one witness extraction), the monitoring module (safe and common safe-action sets, the partition-form adequacy converse, maximal common-action sets, the partition census, the review-timing identity with its threshold convention, and the certainty-equivalence drift audit), and a command-line self-test of twelve audited identities — the four-cell kernel counts $24/26/25/28$ with the meet identity, the census 15 to-

tal and 7 adequate partitions with maximal sets overlapping in $(2, 2)$, the benchmark $Y^* = 27/5$, the drift bounds $21/100$ and 0 , the 48-cell delay identity with its boundary cell, and witness extraction — each a pointer to the published audits (Abaee, 2026, An obstruction calculus; Abaee, 2026, Worked systems for the obstruction calculus). The inherited counts are software self-tests pointing to the published audits; they are not used in this paper’s main theorem or example, and their full context lives in the package documentation. In the certification workflow the library plays the enumerate-and-check role: finite instances are enumerated exactly, candidate designs are tested against the fibre criteria, and every claim is archived with the script that regenerates it. The library is not a reimplement of this paper’s bridge theorems: it carries the exact finite machinery they reduce from, while the bridge’s programs live in the campaign scripts of Section 14; consolidating those scripts as a `viacert.continuous` module is the library’s recorded next step, together with three recorded tooling directions: warm-started incremental refinement (re-solve on block splits seeded with the coarse dual, benchmarked against cold re-solve), an arbitrary-precision dual re-solve fallback for interval-coefficient programs (solve at working precision, detect dual-gap tightness, re-solve the tight dual in extended precision), and conflict-targeted block refinement where kernels are non-affine. Consolidation supersedes implementation paths only in future releases: the archived scripts and their audited identities remain the exact versions used for this article’s verification record. The title’s “reference library” is read as this finite-side package plus the certified campaign pipeline.

12. Scope delimitations

The exact theorem does not cover: control-dependent observations or active sensing; arbitrary continuous-state measurements; nonlinear dynamics without additional validated enclosures; nonconvex actuator sets without a relaxation gap (convexity is structural: block averages of admissible controls lie in $\text{conv } U$, and the weak-* compactness of Proposition 2 uses that U is a compact convex polytope); rate-constrained actuators without modifying the approximation; or unrestricted infinite-horizon completeness. For general sensing after a blackout, the oracle-recourse certificate is sound but incomplete — exactness in the audited instance belongs to the complete revelation at τ , not to the relaxation. For nonlinear systems, a uniform validated flow-error bound can be added to the row budgets, yielding a sound extension, not automatic completeness. Against the alternatives: full-information viability is necessary but not sufficient for the information-constrained question; initial-admissibility and tangency tests are silent on the instance’s obstruction, which is temporal-informational rather than pointwise-geometric; safety-before-first-observation tests are silent when a common safe blind control exists; exact finite safety games exist as discrete models, and what is needed to use them as sound abstractions of the present continuous systems is a justified continuous-to-finite bridge, which is what Theorem 4 provides; and naïve held-control infeasibility excludes only the held-control class. No general-purpose computational-calculus claim is made: the theorem is a

certificate generator for the stated class, with completeness only under refinement and semidecidability only with effectively computable bounds.

13. Conclusion

The general finding first, because it is the one that survives the instance. The cost of certifying a system under incomplete observation is not governed by the size of the system, and it is worth saying precisely what it *is* governed by, because the two are routinely conflated. The certificate produced by the pipeline of this paper has a size that scales with the *information–time product*: the product of how much the controller can learn and over how long a horizon it must act. It does not scale with the input dimension — the dimension of the state, of the disturbance, or of the discretisation standing in for either. A high-dimensional system about which little can be learned over a short horizon is cheap to certify; a low-dimensional system observed richly over a long horizon is expensive. Dimensionality is the wrong axis, and pipelines organised around reducing it are optimising the wrong quantity. This is the paper’s transferable claim, and it is a claim about the *shape* of the problem rather than about any instance of it. The same conclusion is reached independently, in a different language and for a different certificate form, in the measure-dual companion (Abaee, 2026, minimax dual certificates), where the witness size is set by the control dimension k with a tight $k + 1$ bound. Two different constructions, one moral: *obstruction certificates are small objects whose size is fixed by the decision and information structure, not by the size of the world.*

What that claim rests on. A sound and independently checkable certification pipeline under incomplete observation rests on three requirements, and the bridge theorem delivers all three. First, the finite controller class must be an *outer* relaxation while the stored adversarial tests correspond to genuinely admissible scenarios — relax the controller, not the adversary, or the certificate certifies nothing. Second, optimality must be certifiable independently of the solver: the solver’s output is a confirmation, not a proof object, and the exact witnesses are what close the gap. Third, and this is the requirement above, certificate size must be understood in the information–time product rather than the input dimension.

What the instance shows. The three-branch instance shows the resulting certificates detecting failure that full-information viability, blind-window exit tests, and held-control infeasibility all miss, and the campaign shows the solver returning the independently proved optima to numerical precision while exact witnesses close every gap. The reference library carries the exact finite machinery to new systems.

What remains open. Completeness beyond refinement: an inner approximation of the policy class that is computationally effective for general sensing is not known, and the oracle-recourse extension is the current sound surrogate.

Three further extension directions are recorded, each a construction of its own rather than an increment of the present framework; all lie outside the audited class delimited in Section 12. The cell-conditioned lift of labels, kernels, and costs for general measurement channels — the

passage from history partitions to general measurement maps — is carried in Proposition 6, with the instance’s compatible-mode hierarchy and radius bounds as the audited instantiation. (i) Stochastic disturbances: replacing support-function rows with second-moment bounds turns the finite program into a conic one (second-order cone or semidefinite), with exact rational witnesses replaced by conic dual certificates; the exact rational optima of the present campaign remain the benchmark such relaxations must meet. (ii) Decentralized and networked controller sets, infinite-horizon and periodic regimes, and nonconvex or piecewise-smooth actuators: each alters a structural input of the bridge theorem (the weak-* compactness of label strategies, finite-horizon completeness, or the convexity of the actuator set), and is therefore a new construction rather than a parameter change. (iii) A cross-tool comparison study that treats the audited instance as a common benchmark — exact finite safety games, interval-reachability toolchains, SMT-backed barrier synthesis — with the certificate file as the exchange format; the tooling directions recorded in Section 11 and the independent certificate-file parser of Section 14 are its computational prerequisites.

14. Verification methods

Every reported mathematical benchmark value and every table entry is regenerated by deterministic scripts, exact rational arithmetic unless stated as solver output, standard library plus the solver interface only.

A mechanized layer, distinct from the scripts. The checks catalogued below are computations; they do not prove. Separately, the Farkas core on which the exact primal–dual witnesses rest is *proved* in a mechanized layer (Lean 4, 60 preserved modules under **Formalizations/**; the default target builds 60/60 jobs and the checked soundness declaration has no admitted gap): the soundness direction — the one the certificates actually use — is derived from the ordered-field interface alone. **Current Lean provenance.** The committed project pin v4.34.1 was rebuilt in GitHub Actions (default Lake target: 60/60 jobs, exit 0; <https://github.com/MIKEAA2020/general-sustainability/actions/runs/36946730855>). The exact project source and two footprint-check `.lean` files are archived with SHA-256 manifests. A comment-aware scan of all 62 source files finds no `sorry`, `admit`, explicit `axiom` or `constant` declaration. Twenty-one named footprints contain no `sorryAx`; the aggregate-import environment exposes three source-level `native_decide` uses in the P3 support/value examples, yielding nine distinct generated axiom names under this pin. These are executable computational certificates that extend kernel-only trust to the Lean compiler/runtime; they are not admitted gaps and must not be described as zero axioms or as kernel-only proofs. Three preserved historical modules outside the aggregate import closure are not certified by the default build. The v4.14.0 comparison of the *current* source fails in the prelude and does not reproduce the old source run. The converse direction is not proved there, and is not used: its classical argument rests on a separation theorem the layer does not develop. Three results of this paper remain out of reach of that layer by construction, and

are cited rather than formalized: the value identity, the bridge theorem, and the belief-cell proposition, each of which calls for analysis (exponentials and logarithms, or weak-* compactness) that an ordered-field interface does not supply. That is a statement about the formalization, not about the results. Three scripts and one package cover the claims, with one master entry point chaining them. The exact verification record re-derives the instance’s certificates — delay threshold, margins, witnesses, erosion identity, rank construction — in 51 exact checks. The campaign script rebuilds the program (5) at each of the five meshes, runs HiGHS, and certifies optimality by the exact primal–dual witnesses, closing the duality gap; it re-derives the mesh law and all dimension identities (nine checks). The library self-test runs the twelve audited identities. A master script chains the three, then recomputes independently: the mesh values from the closed law $3/50 - Th/4$ (uniform aligned meshes), the Farkas contradiction $-\frac{53}{100} + \frac{1}{2} = -\frac{3}{100}$, the primal row value $\frac{3}{100}$, the margins $\frac{3}{50}$ and $\frac{3}{100}$, the moment error $e = \frac{3}{100}$, the relaxed right-hand side $-\frac{53}{100}$, the delay threshold $\tau^* = \frac{7}{50}$, the blind-window safety value $\frac{39}{25}$, and every headline number in the text; independence here means a separate formula path from the subordinate scripts, not a separate implementation — the independent certificate-file parser is shipped (`certificate_exchange_v1.py`) and validates the archived witnesses of this paper and of the dual-certificate companion (Abaee, 2026, Minimax dual certificates) exactly, in either direction, alongside the tooling directions of Section 11. The two figures are generated by `paper2_comp_certification_figures_v1.py` (recompute-then-assert: every plotted peak, threshold, and crossing is recomputed in exact rational arithmetic and asserted against the printed values before the figure is written; ten assertions). Timings are single runs on the verification machine (Linux x86_64, Intel Xeon @ 2.60 GHz, Python 3.13, SciPy 1.17 HiGHS dual-simplex interface, solver defaults otherwise).

1. Related work

This paper builds a finite, solver-attainable, independently checkable certificate for a continuous-time safety value under partial observation. Each of those four properties has an established literature behind it, and this section says where the present construction sits relative to each.

1. Viability kernel computation, and the sandwich

The computation of viability kernels by discrete approximation is the classical route. Saint-Pierre’s viability kernel algorithm (Saint-Pierre, 1994) approximates the finite-horizon kernel by a backward recursion on a grid, with a convergence statement: for open constraint sets, the union of the approximating sets over vanishing time step recovers the kernel. Level-set formulations compute the same objects through Hamilton–Jacobi equations (Mitchell, Bayen and Tomlin, 2005). Because these Eulerian schemes grid the state space, their cost grows exponentially with dimension, which motivated Lagrangian reformulations in terms of reachable sets that scale considerably further (Maidens,

Kaynama, Mitchell, Oishi and Dumont, 2013).

The two-sided structure is not new, and this paper does not claim it is. The reachability-based viability computations produce a guaranteed *under*-approximation of the kernel together with a free *over*-approximation, and use the latter to bound the error of the former (Maidens et al., 2013, Section 3). The present paper’s sandwich,

$$\rho_{\mathcal{H},G} \leq J_{\mathcal{I}}(T) \leq \rho_{\mathcal{H},G} + \bar{\epsilon} + \delta_{\beta} + Lh_t,$$

is of that kind: a computable quantity, and a certified inflation that bounds the true value from the other side. What differs is the object being sandwiched and the approximations that generate it. The classical schemes discretise a *state space* and approximate a *kernel*, a set of states. Here the discretisation is over an outer moment relaxation of the admissible *information-adapted controls* together with an inner adversarial selection of finitely many stored labels and scenarios, and the quantity bounded is a safety *value* over an information state — not a set of initial conditions. The partial-observation structure, and the fact that the certificate must hold against every measurable policy rather than against one synthesised controller, is what forces that change.

2. Moment relaxations

Relaxing a hard optimisation problem into a sequence of convex problems whose optimal values converge monotonically to the true optimum is the moment–SOS programme (Lasserre, 2001; Parrilo, 2003), whose hierarchy produces nondecreasing lower bounds with asymptotic convergence and, importantly, *checkable* conditions — rank and flat-extension criteria on the moment matrix — under which a relaxation is certified exact and minimisers can be extracted. That certification idea is the direct ancestor of the present one.

The differences are concrete. The moment–SOS hierarchy applies to polynomial optimisation over semi-algebraic sets and is solved by *semidefinite* programming; the relaxations here are *linear* programs over moment variables for a linear system, and the programs are solved by a standard LP solver. Convergence in the hierarchy is in the relaxation degree; convergence here is along a refinement sequence whose mesh and stored-scenario and enclosure widths vanish (Theorem 4). And the certified object differs: an exactness certificate in the moment–SOS setting licenses the extraction of a global minimiser, whereas here a strictly positive certified margin is an *obstruction* certificate — a proof of nonviability, not the recovery of an optimum.

3. The scenario approach, and why the guarantee here is not probabilistic

Sampling a finite set of uncertainty instances and solving the resulting finite program is the scenario approach (Calafiore and Campi, 2005, 2006), with sharp sample-complexity results (Campi and Garatti, 2008; Campi and Garatti, 2011, on sampling-and-discarding; Campi, Garatti and Prandini, 2009, for the systems-and-control setting). The inner approximation here — finitely many

stored admissible labels and scenarios — is superficially the same move.

It is not the same guarantee. The scenario approach is *probabilistic*: for N sampled constraints the returned solution violates the unsampled constraints with probability at most ϵ , with confidence $1 - \beta$, where N is chosen to deliver that pair. The guarantee here is *worst-case*: the scenarios are selected adversarially rather than sampled, and the bound holds for every measurable information-adapted policy, with no measure on the uncertainty and no confidence parameter. A reader who imports intuitions from the scenario literature should note that ϵ and β have no counterpart here; the error terms $\bar{\epsilon}$, δ_{β} and Lh_t are deterministic inflation certificates, not violation probabilities. (The notational collision on β is unfortunate and is flagged rather than resolved.)

4. Verified numerics, and the role of the solver

Enclosing the solution of a continuous problem in rigorously bounded intervals is the programme of interval arithmetic and validated ODE integration. This paper deliberately does not use it. Every exact verification path for the reported rational benchmark runs in integer and rational arithmetic, and optimality is certified by independent primal and dual witnesses checked exactly. **The solver is not a proof object**: the numerical optimum returned by the LP solver is a confirmation that agrees with the exact rational value to within 4.2×10^{-17} on the reported cases, and the certificate rests on the exact witnesses, not on the floating-point output. This is a statement about the division of labour, not a claim that interval methods are inferior; they address problems this pipeline does not.

5. What is claimed

Given the above, the contribution is five things.

1. **A continuous-to-finite bridge for information states**: an outer moment relaxation and an inner adversarial discretisation combine into a finite LP whose value and whose error inflation bound the continuous-time safety value on both sides, with convergence along any admissible refinement sequence (Theorem 4).
2. **Dual-feasibility as a universal certificate**: every dual-feasible solution of the finite program yields a finite lower-bound certificate covering *every* measurable policy, and a strictly positive certified margin is a finite obstruction certificate.
3. **The witness count is governed by information–time rank, not input dimension**: whenever a positive certificate exists one can be chosen with at most $r + 1$ safety labels at rank r , and a matching construction with scalar input produces obstructions whose minimal certificates require arbitrarily many witnesses, saturating the bound (Proposition 5). The negative half — that a single input does not bound the witness count — is the part most likely to surprise.

4. **The certificate is prescriptive**, decoding into an exact shared-authority hierarchy and into redesign sensitivities that separate what buys tolerance from what buys nothing (Propositions 7 and 8).
5. **Exact verification paths by construction**: primal and dual witnesses checked in rational arithmetic, with the solver demoted to a numerical confirmation.

Not claimed: the sandwich idea as such (Maidens et al. and the reachability literature), moment relaxation as such (Lasserre; Parrilo), scenario discretisation as such (Calafiore and Campi; Campi and Garatti), or interval enclosure. The novelty, such as it is, lies in assembling these into a certificate that is valid against every measurable information-adapted policy under partial observation, and in the rank result of item 3.

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Supplementary material

The complete proofs and verification record cited in the bridge theorem accompany this article as `paper03_computational_certification_v16_supplementary.tex` (with a matching PDF). The companion’s mathematical text has not been rewritten here; its source-year identity is recorded in the preservation review.

Declarations

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Competing interests

None.

Data availability

No external data were used.

Code availability

The verification scripts `paper2_computational_certification_v20_verification.py`, `paper2_nonstandard_verification.py`, `paper2c_lp_campaign.py`, and the `viacert` package are publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`, at the commit pinned to this article’s submission); they reproduce all exact values, witnesses, bounds, and the solver campaign verbatim.

AI declaration

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited outputs and takes responsibility for the final work.